



Setting Up WiFi At Home

Here's all you need to know about WiFi routers

Robert Sovereign Smith

With an increasing number of us buying laptops—coupled with more broadband connections—we've seen a spike in the number of multiple-PC households. It's quite common—especially in metros and A and B class cities across India—to find a PC plus a laptop or two (and even WiFi-enabled PDAs) in the average tech-savvy household. The need for networking them arises, however, and WiFi routers seem to be the popular choice. This is where the tech-savviness ends, sadly, with more unsecured wireless networks popping up all the time. Here we'll walk you through some important aspects of setting up and managing your network.

What's Popular

If you have a broadband connection that requires an ADSL modem, you should consider buying an ADSL router. These have the modem inbuilt, and also act as a router for your network. If your ISP has already provided you with a modem, you'll only need the average WiFi router. We've found the Linksys WRH54G router to be the most popular buy, and it's what we've used as an example here. The slightly older Linksys WRT54G is also a very popular model; the only difference is in the looks, and the fact that the (newer) WRH54G has a Wizard page in the control panel to help you set things up easily at home. Routers from D-Link, NetGear and others are just



The Linksys WRH54G, for home users

as good—just the terminology and nomenclature of the settings we mention in the following will differ from what you find on your screen if you own a different brand of router.

Setting It Up Right

It's best to let your router share your Net connection, so that the laptops or other devices on the network can access the Internet even when the desktop PC (if you have one) is switched off. If you buy an ADSL router, you will find that your ISP's cable fits directly into it. If you already have a modem, and if you've bought a WiFi router, you'll see that it has LAN ports at the back. One of these ports will be called "Internet" or "WAN", and this is where you connect the LAN cable between your router and the modem.

Router Settings

All the routers available use a browser-based control panel, which you access by typing in a specific address into the address bar. The first time you hook up your router to your PC or laptop, you should use a LAN cable (usually bundled) and connect it to one of the router's LAN ports. Set the Local Area Network connection on your PC / laptop to obtain an IP automatically. To do this, go to Start > Settings > Network Connections. Right-click on your LAN connection and select Properties. Under the "This connection uses the following items:" header, you should see Internet Protocol (TCP/IP). Double-click this and make sure the Obtain an IP address automatically radio button is selected. Click OK and then, back in the LAN Properties box, make sure to check the Show icon in notification area when connected box at the bottom before clicking OK again.

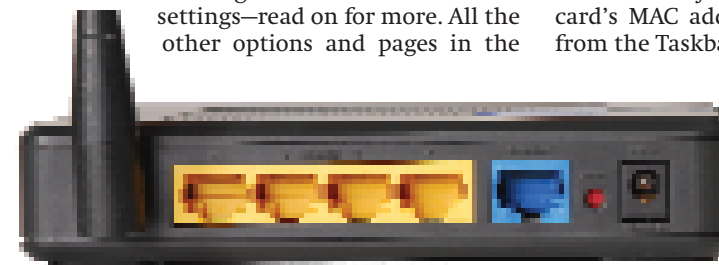
Usually, the documentation that accompanies the router will tell you what address to use

to access the control panel, but if you've have lost the documentation, you can just wait for the LAN connection (between the router and the PC / laptop) to get running, and then, once you have been assigned an IP address, look at the details and note down the "Gateway" IP. You should see the LAN icon in the Taskbar when the connection is available; just double-click on it and select the Support tab. Here you will see values for IP, Subnet Mask and Default Gateway. Type the Gateway IP into your browser and you should get into the control panel. You might have to use "admin" as the username and a blank password if prompted. Your manual will have this information, but if you've lost it, look up the manufacturer's Web site.

The Wizard Page

If you're using the WRH54G, you will be shown a Wizard page that will let you set up your router quickly and easily with very well-explained settings. All this should be self-explanatory, and you should have no trouble.

You might need to tweak some settings—read on for more. All the other options and pages in the



The router has clearly demarcated ports for LAN and your Internet connection

WRH54G and WRT54G are identical.

The Basic Setup

If you're using the WRT54G router, the first screen you'll see is the Basic Setup configuration options. Here you specify details of your Internet connection and LAN and wireless settings. Other routers will have similar settings. Important settings to remember are:

Internet Connection: Here you will be asked for the connection type (usually PPPoE for ADSL in India), and the username and password that your ISP should have provided you with. Enter them in the appropriate boxes.

Network: Here you get to choose what IPs your LAN should use, and what IP the gateway should use. You should use private IP addresses here, such as 192.168.0.1, 192.168.100.1,

10.0.0.1, etc.

Servers: You can set whether the router assigns IPs dynamically to clients that connect to it, and which address to start assigning from. In our example, the router is set to use DHCP, and starts assigning IPs from 192.168.100.100.

Cloning

Some ISPs lock user accounts to their network card's MAC (Media Access Control) address. This is to ensure that no one but you or the PC that the connection is registered to is allowed to use the Internet connection. Some ISPs even lock the username and password, so that you can only access the Net from your own home, using only the LAN card that the connection was set up on. If you find that your router is not able to dial and establish a connection with your ISP's server, this could be the reason.

Most routers will let you just spoof your network card's MAC address so that the ISP will let you in. To do this, just click on MAC Address Clone (for WRT54G) and set the MAC address to the value of your LAN card. To find your LAN card's MAC address, double-click on the icon from the Taskbar (or go to Start > Settings

> Network Connections, and double-click your LAN connection), go to the Support tab and click the Details button. You will see a value for "Physical Address"—this is the LAN card's MAC address. Copy it and then paste it into the appropriate place

under MAC Address Clone in the router control panel. You should now be able to dial up and connect to your broadband connection.

Wireless Settings

Under Basic Wireless Settings, select "Mixed" for Wireless network mode if you have multiple devices connecting—laptops, media servers, smart phones, PDAs, Sony PSP, etc. If you have just a laptop and you know for a fact that it supports 802.11g, you can safely set this to "G-Only". Apart from better speeds and connection strength, this will also give you a slight security edge, as people with older WiFi equipment (802.11b) will not be able to connect.

You will also see an option for SSID (Service Set Identifier). Set this to anything

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